

Operators

an **operator** is a symbol or keyword used to perform operations on values

1.Arithmetic Operators

Operator	Description	Example	Result
+	Addition	5 + 3	8
-	Subtraction	5 - 3	2
*	Multiplication	5 * 3	15
/	Division	5 / 2	2.5
%	Modulus (remainder)	5 % 2	1
**	Exponentiation	2 ** 3	8
++	Increment	let x = 5; x++;	6
--	Decrement	let x = 5; x--;	4

Tricky part:

++ and -- can be **prefix** or **postfix**, and they behave differently:

let x = 5;

console.log(x++); // prints 5, then x becomes 6

console.log(++x); // x becomes 7, then prints 7

2. Assignment Operators

Operator	Example	Meaning
=	x = 5	Assign 5 to x
+=	x += 3	x = x + 3
-=	x -= 2	x = x - 2
*=	x *= 4	x = x * 4
/=	x /= 2	x = x / 2
%=	x %= 3	x = x % 3
**=	x **= 3	x = x ** 3

3. Comparison Operators

Operator	Example	Result
==	5 == '5'	true (type coercion!)
===	5 === '5'	false (no type coercion)
!=	5 != '5'	false
!==	5 !== '5'	true
>	5 > 3	true
<	5 < 3	false
>=	5 >= 5	true
<=	5 <= 4	false

== performs **type coercion**, which can be confusing.

```
console.log(0 == false); // true
```

```
console.log(0 === false); // false
```

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/=	x /= 2	x = x / 2
%=	x %= 3	x = x % 3
**=	x **= 3	x = x ** 3

4. Logical Operators

Used with boolean values (true/false).

Operator	Example	Result
&&	true && false	false
,		,
!	!true	false

Short-circuiting:

- && stops evaluating as soon as it finds false.
- || stops evaluating as soon as it finds true.

```
let a = 0;
```

```
let b = 10;
```

```
console.log(a && b); // 0 (because first falsey value)
```

`console.log(a || b); // 10` (because first truthy value)

5. Operator Precedence

Operators have **priority**, which affects how expressions are evaluated:

`let x = 3 + 4 * 2; // multiplication first → 3 + 8 = 11`

- Parentheses () have the **highest precedence**.
- Multiplication/division > Addition/subtraction > Assignment